

Predicting Dengue Disease Spread

Part 2. Machine Learning Algorithm

Quick Recap

Climatological data

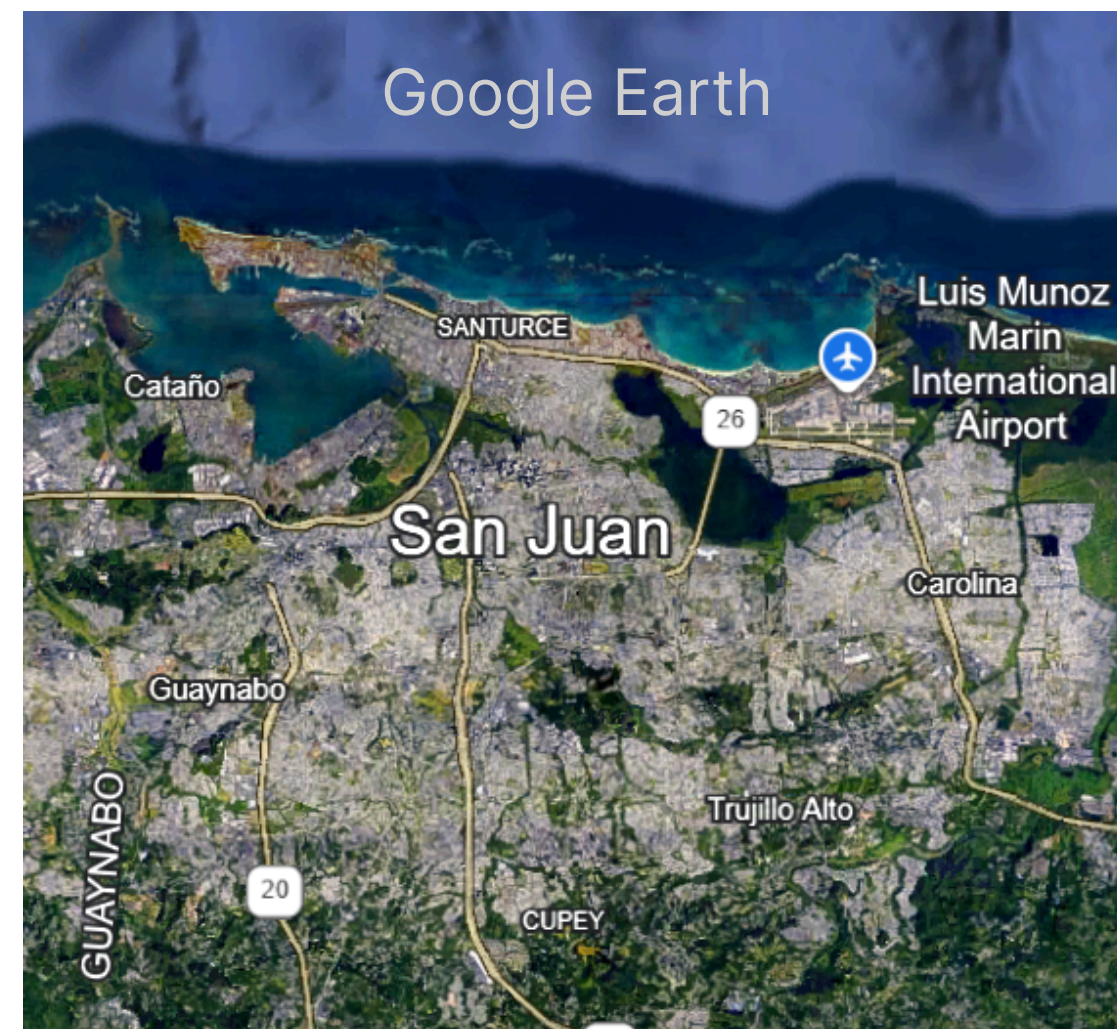
Goal: The dengue disease is carried by mosquitoes that depend on temperature and precipitation. The goal is to predict number of cases of dengue using climate data in San Juan, Puerto Rico.

Preprocessing: Interpolation → Lagged Features → Log Transform → **Feature Engineering (new)**

Hyperparameters Tuning: RollingCV

ML Models: Random Forest Regressor & XGBoost

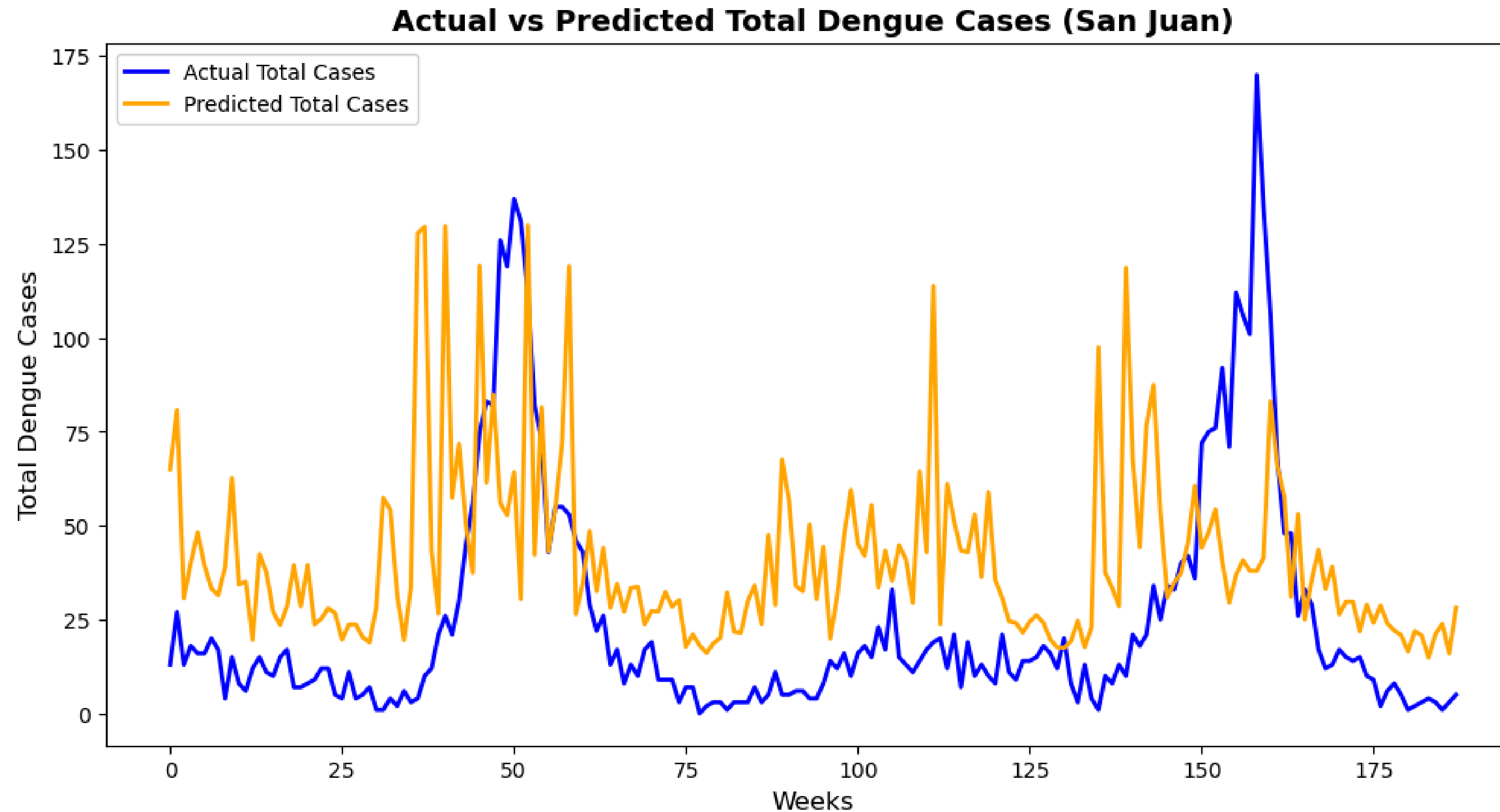
Evaluation Metrics: Mean Absolute Error (MAE) & Root Mean Squared Error (RMSE)



Preprocessing

Experiments 1: Base Model* + **No Preprocessing**

The base model that we're using is *RandomForestRegressor* with default parameters.



Train MAE: **10.46**

Train RMSE: **17.76**

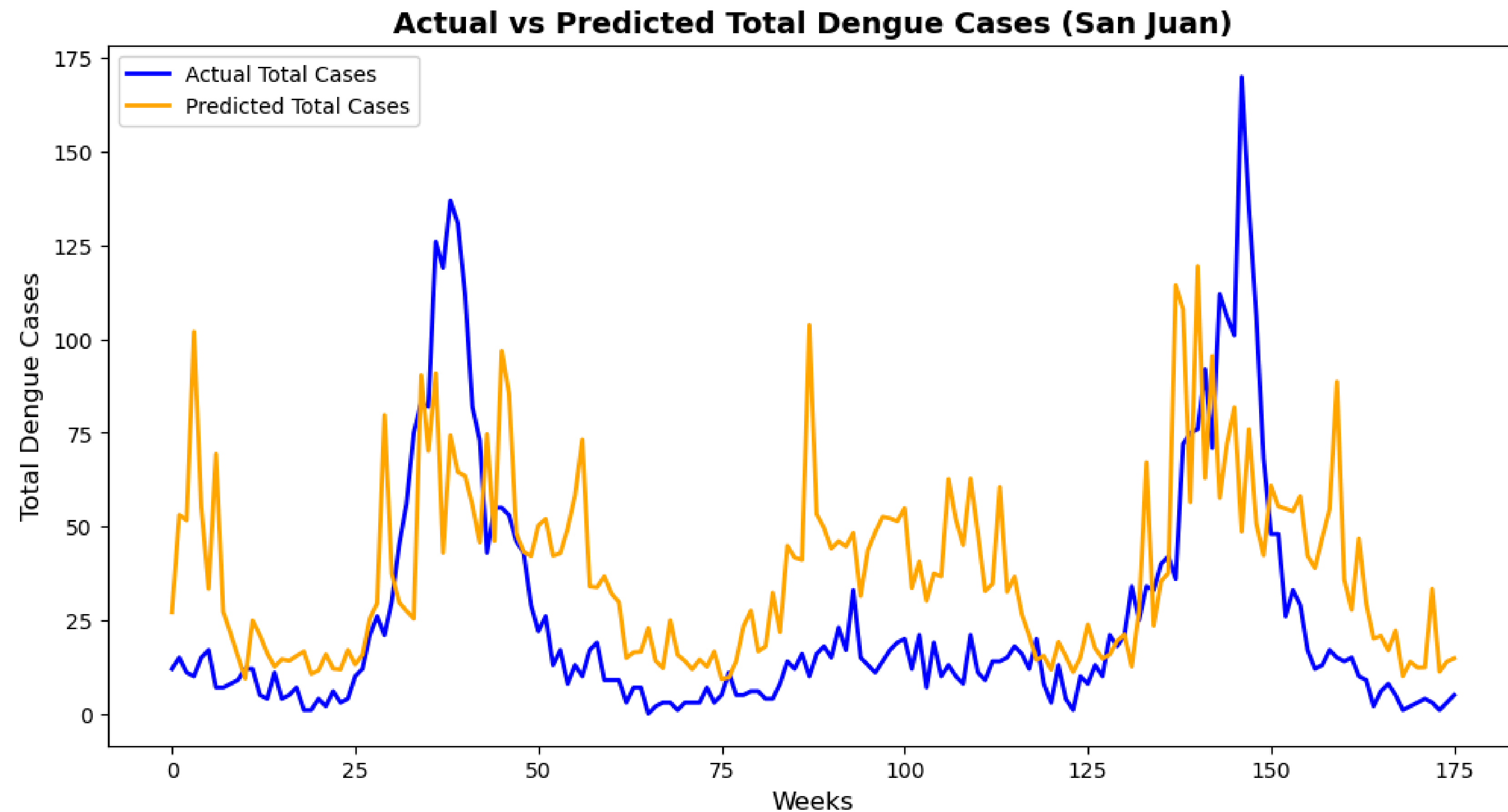
Test MAE: **27.96**

Test RMSE: **36.81**

Preprocessing

Experiments 2: Base Model* + **Lagged Features**

Since dengue transmission is not instantaneous, we shifted the data by amount of lag=10.



Train MAE: **9.60**

Train RMSE: **17.82**

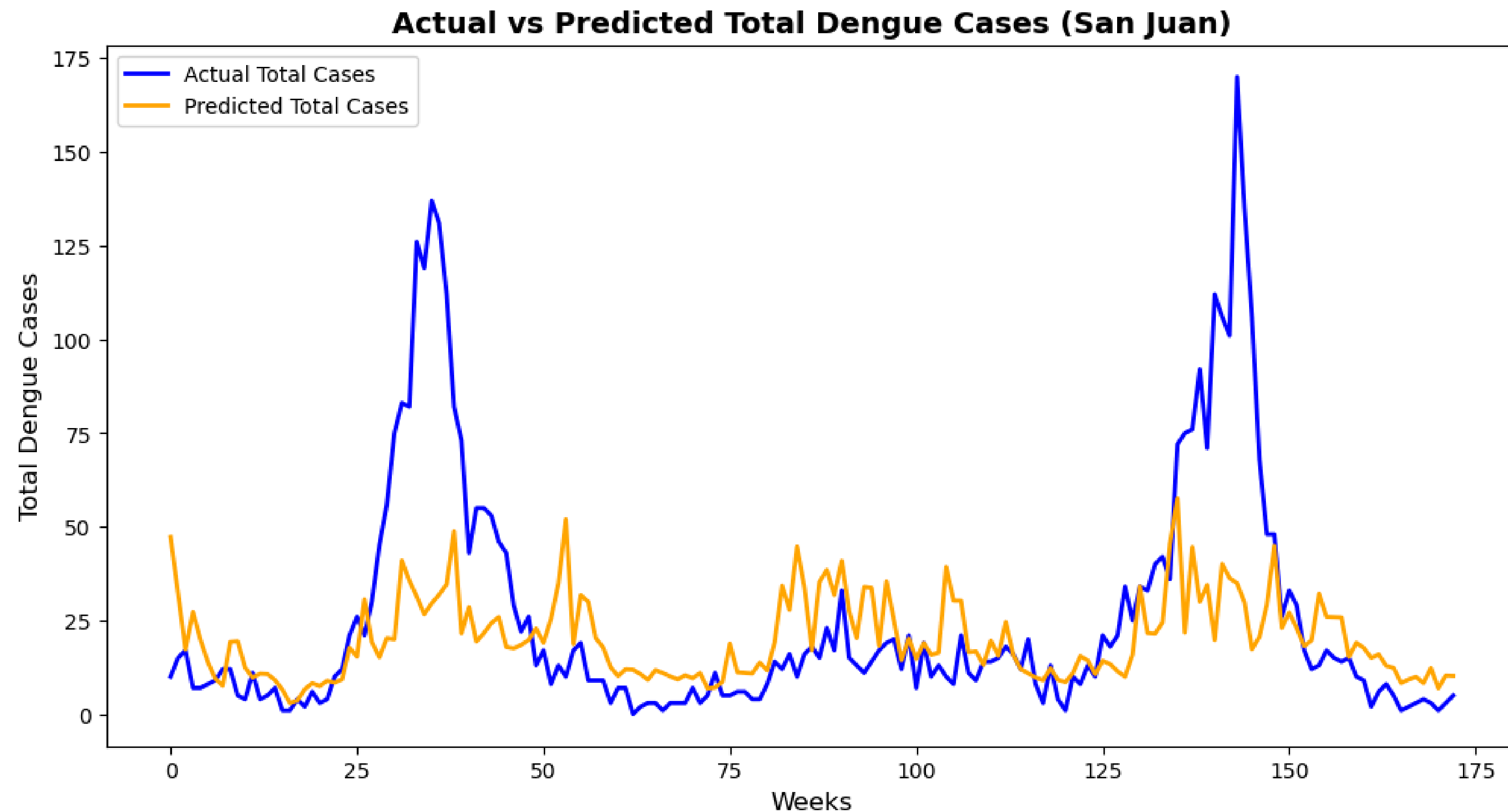
Test MAE: **23.55**

Test RMSE: **30.46**

Preprocessing

Experiments 3: Base Model* + Lagged Features + **log-Transform**

In this process we did log transform to the target variable given the skewed label distribution.



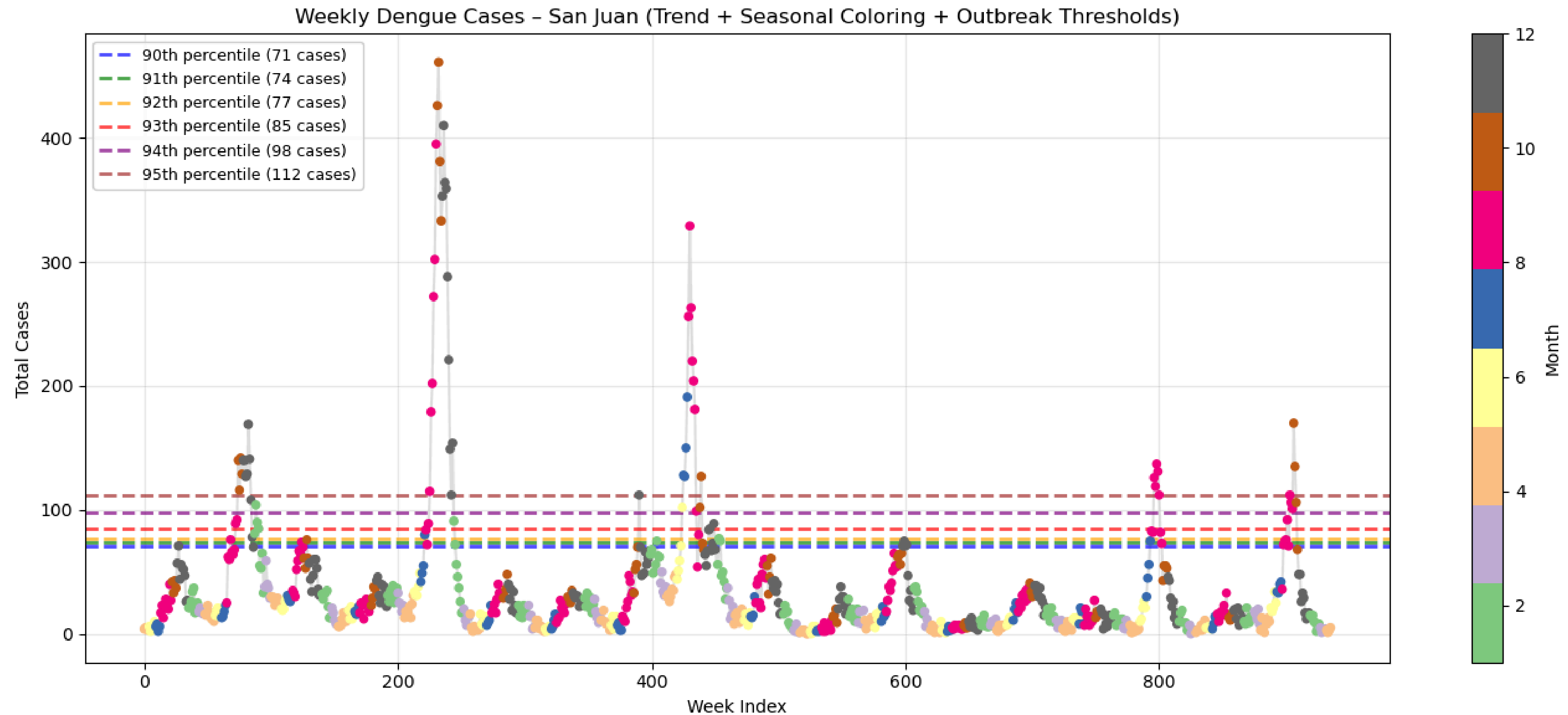
Train MAE: **10.47**
Train RMSE: **29.06**

Test MAE: **17.05**
Test RMSE: **28.98**

Preprocessing

Experiments 4: Base Model* + Lag + Interpolation + log-Transform + **Feature Engineering**

The final step is conducting feature engineering to help the model have the knowledge to detect the spike or in this dataset context is the **outbreak** period.



Preprocessing

Experiments 4: Base Model* + Lag + Interpolation + log-Transform + **Feature Engineering**

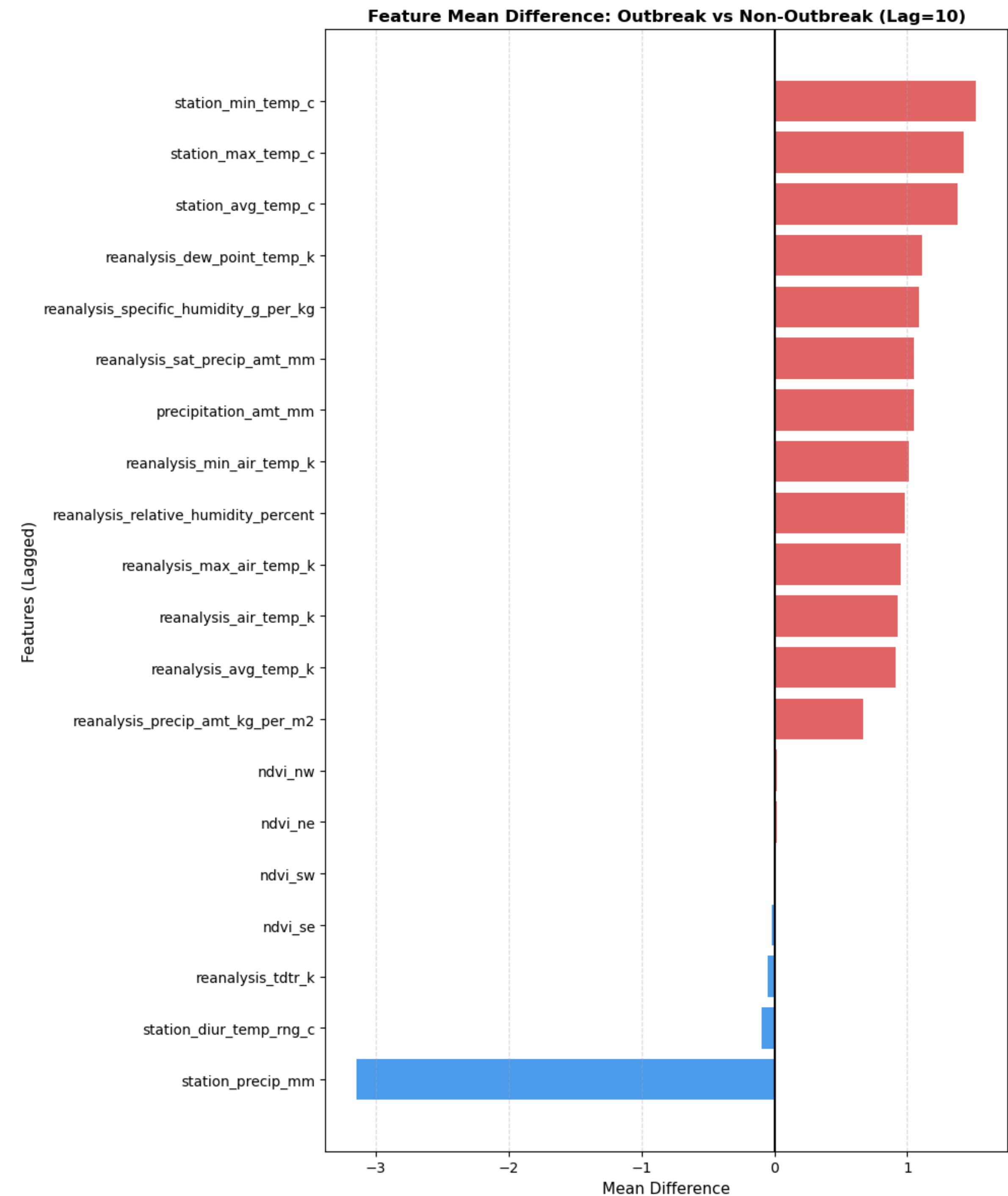
$$\text{precip_roll_mean}_t^{(w)} = \frac{1}{w} \sum_{i=t-w+1}^t x_i$$

$$\text{precip_roll_std}_t^{(w)} = \sqrt{\frac{1}{w} \sum_{i=t-w+1}^t (x_i - \text{precip_roll_mean}_t^{(w)})^2}$$

$$\text{precip_roll_min}_t^{(w)} = \min(\mathcal{W}_t)$$

$$\text{precip_roll_max}_t^{(w)} = \max(\mathcal{W}_t)$$

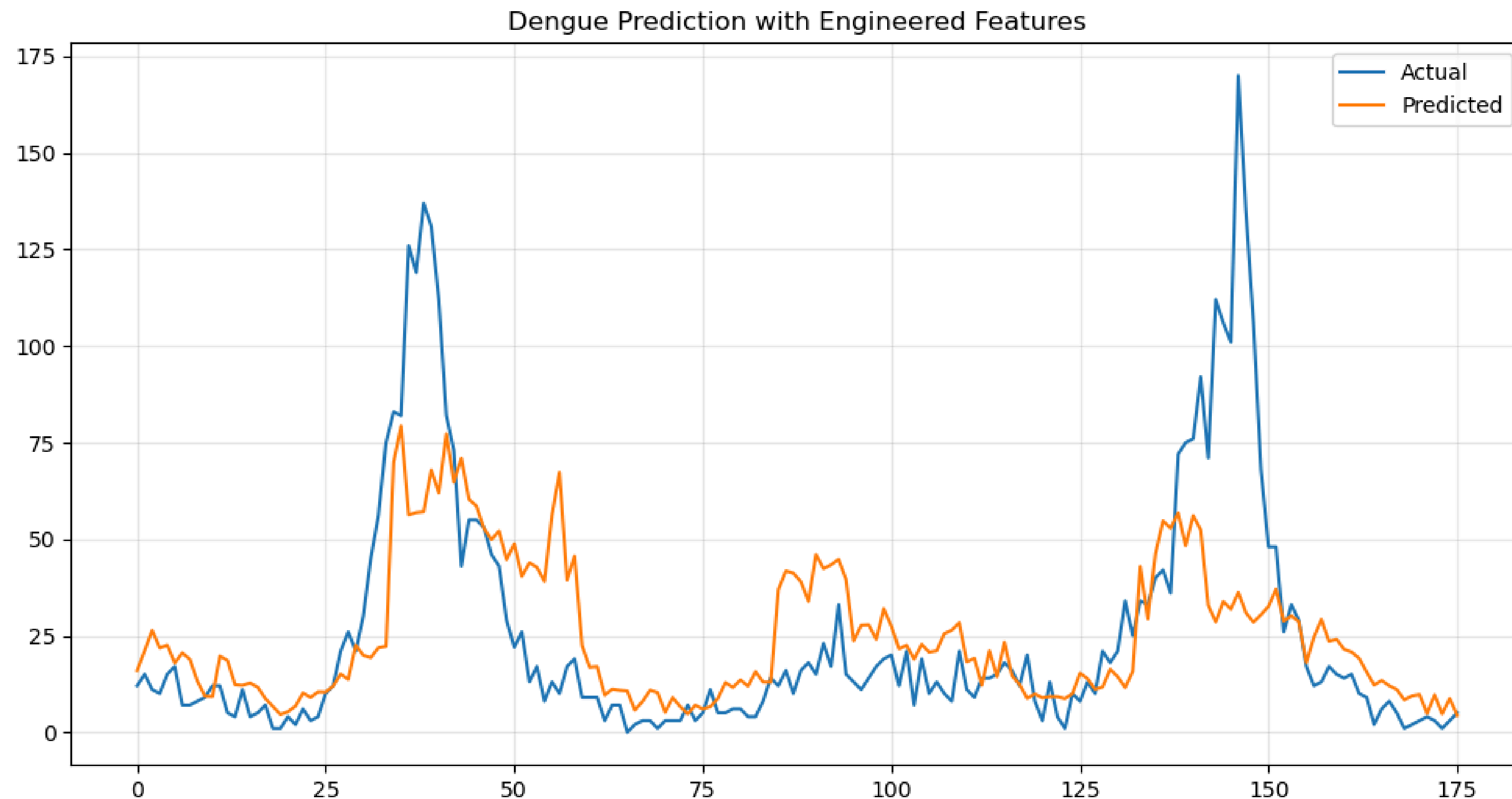
$$\text{precip_anomaly}_t^{(w)} = x_t - \text{precip_roll_mean}_t^{(w)}$$



Preprocessing

Experiments 4: Base Model* + Lag + Interpolation + log-Transform + **Feature Engineering**

After identifying the feature with a high difference value for outbreak periods, we added new features.



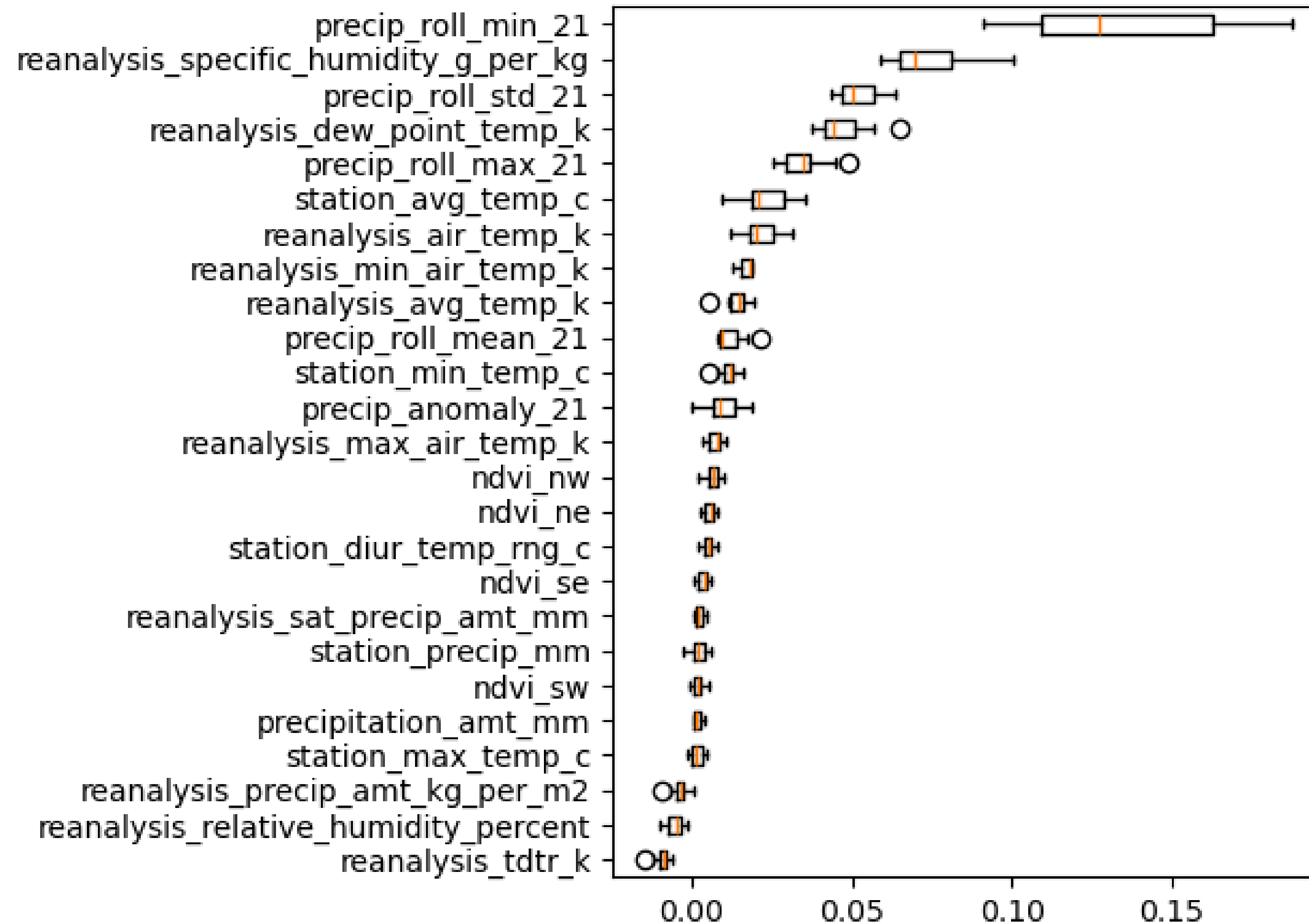
Train MAE: **7.09**
Train RMSE: **20.03**

Test MAE: **14.83**
Test RMSE: **24.89**

Feature Analysis

Mean Decrease Accuracy

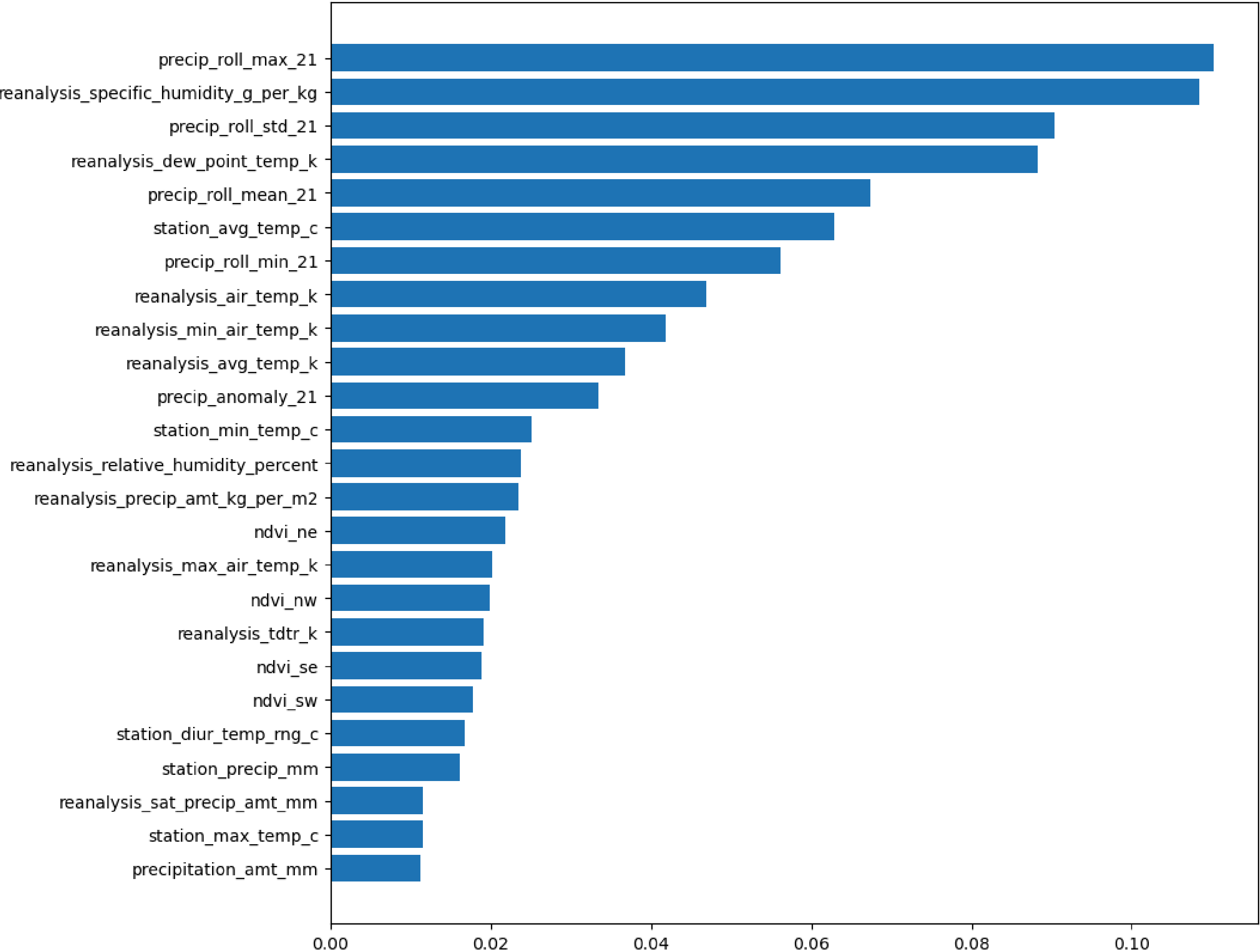
MDA computed on the test set for the 25 features with the highest average MDA (average MDA is displayed in orange)



Feature Analysis

Feature Importances

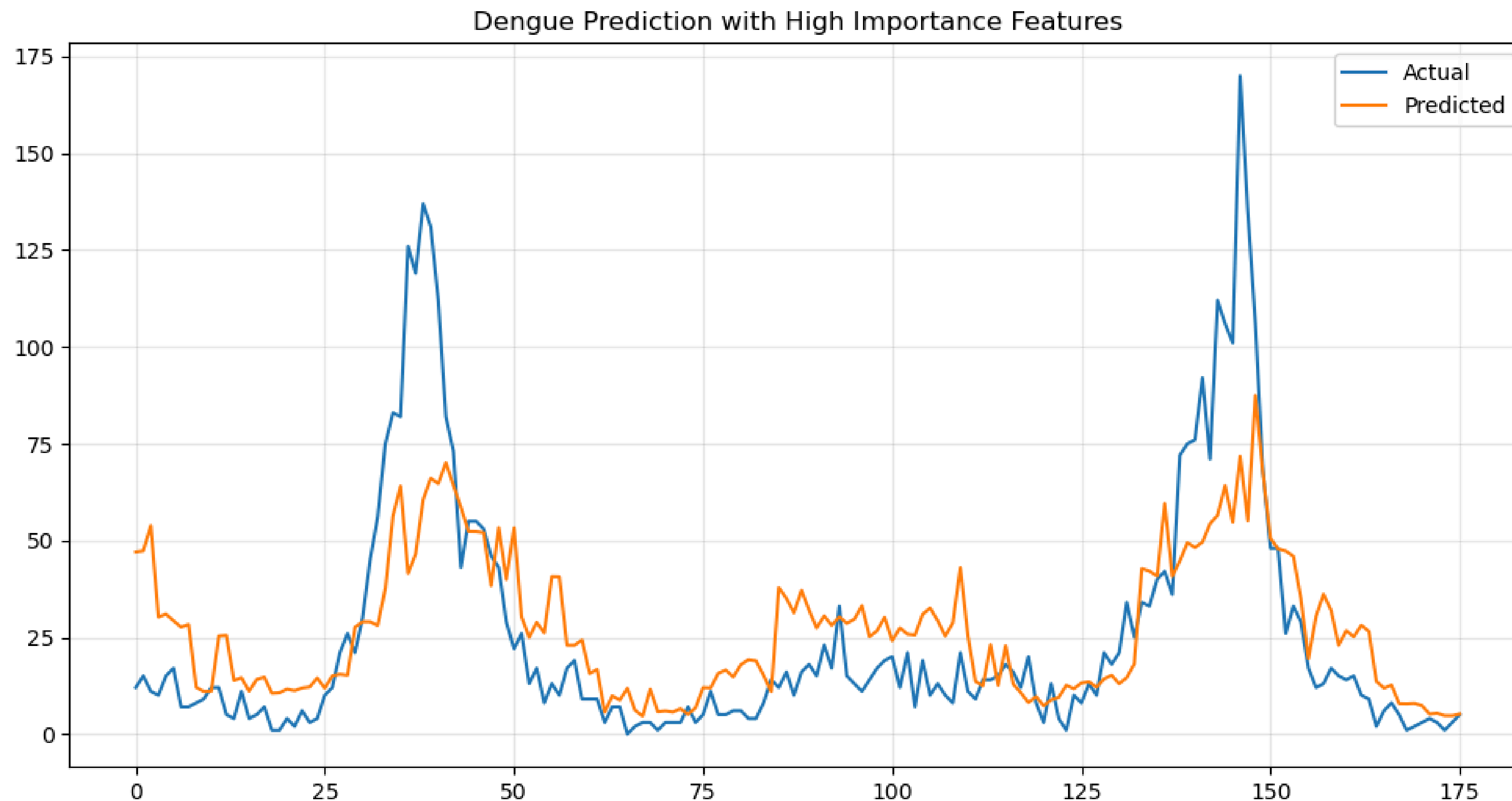
After empirical tuning, we identified the top five features that consistently contributed the most to predictive performance.



Feature Analysis

Base Model* + Preprocessing + **High Importance Features**

After training the model with the previous preprocessing, we retrained the model with fewer features.



Train MAE: **6.96**
Train RMSE: **20.54**

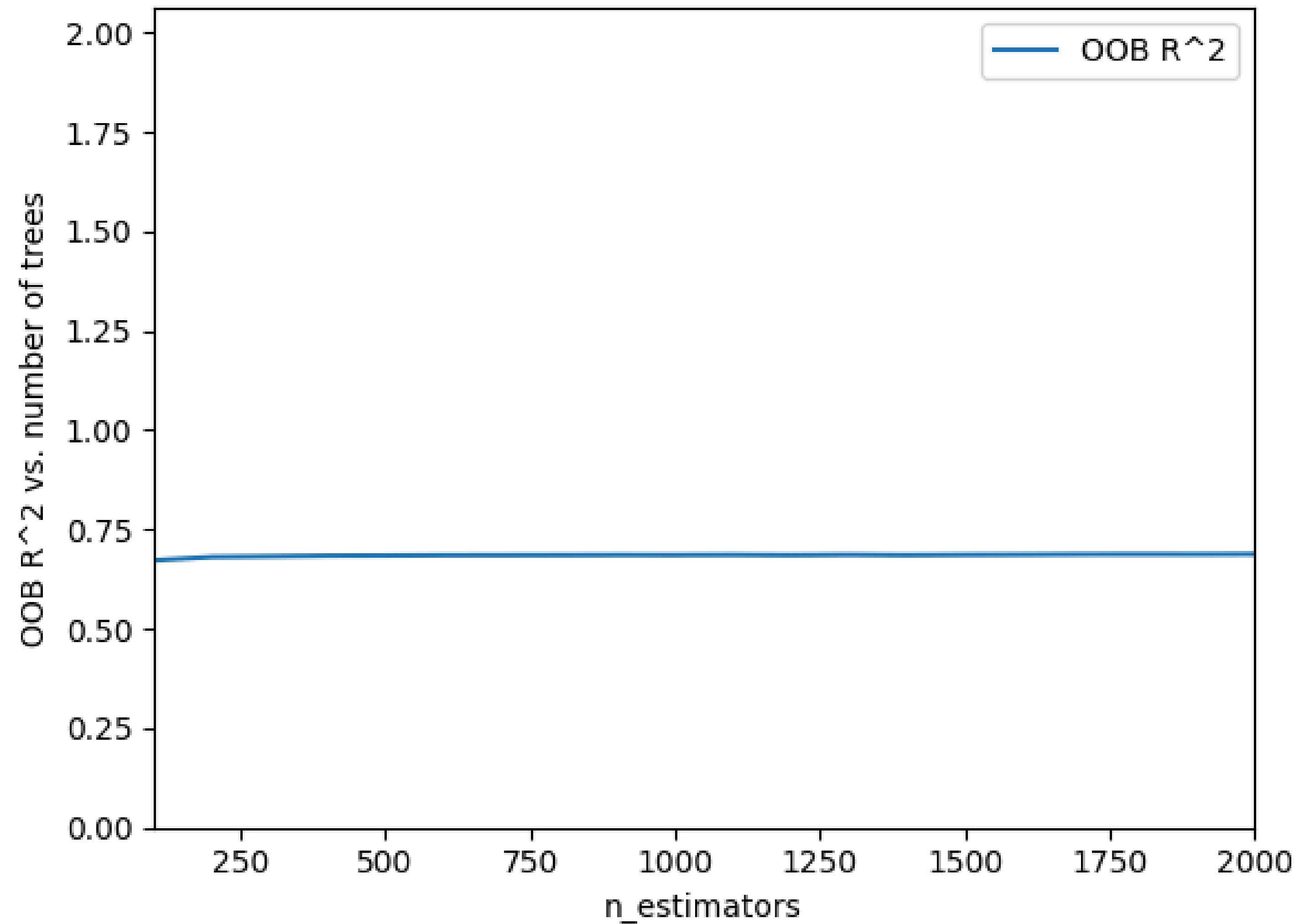
Test MAE: **14.17**
Test RMSE: **21.48**

Hyperparameters Tuning

on RandomForestRegressor

Number of Estimators

Selected: **500**

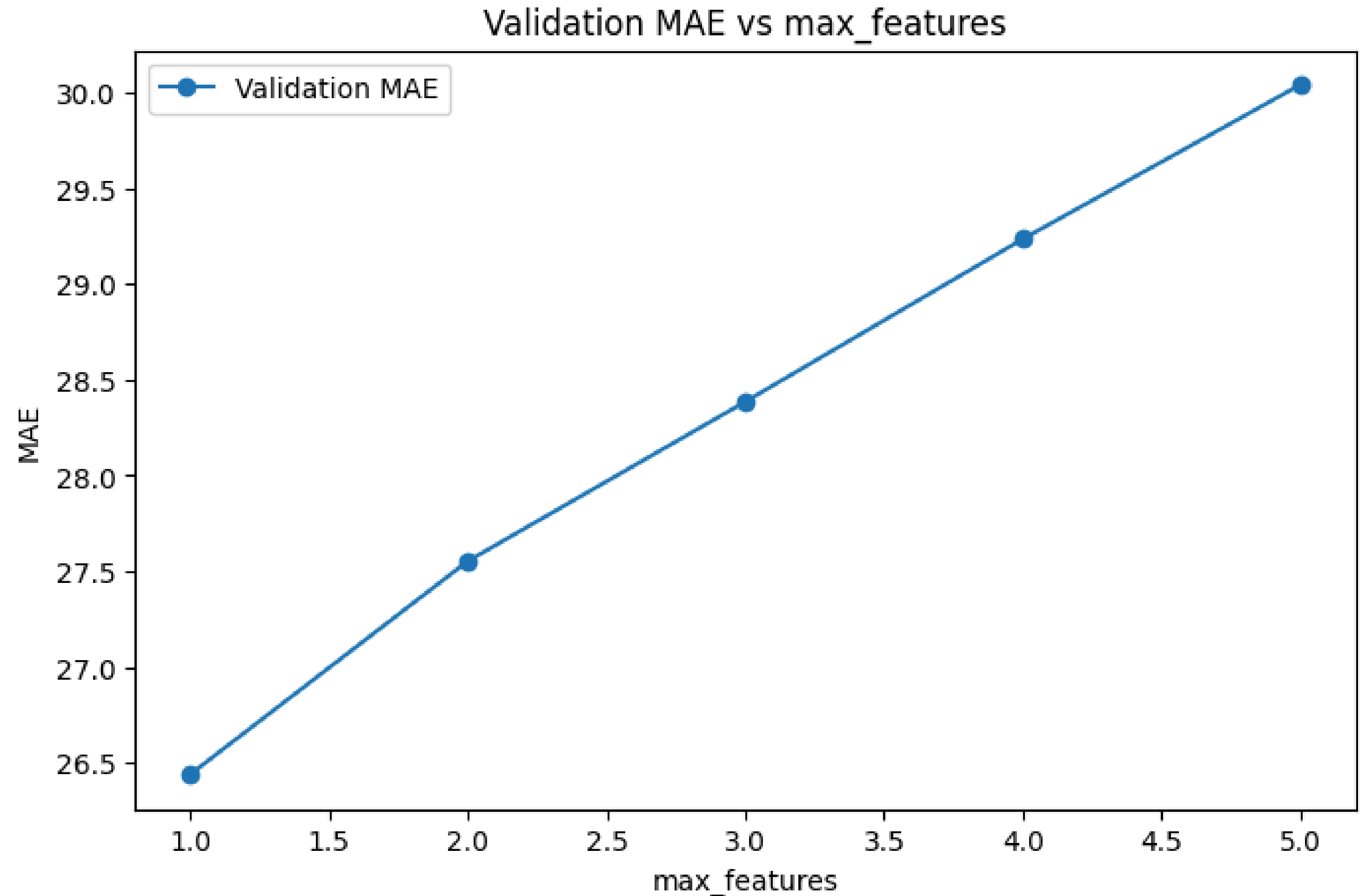


Hyperparameters Tuning

on RandomForestRegressor

Max features

Best: **1** or **n_features/3**

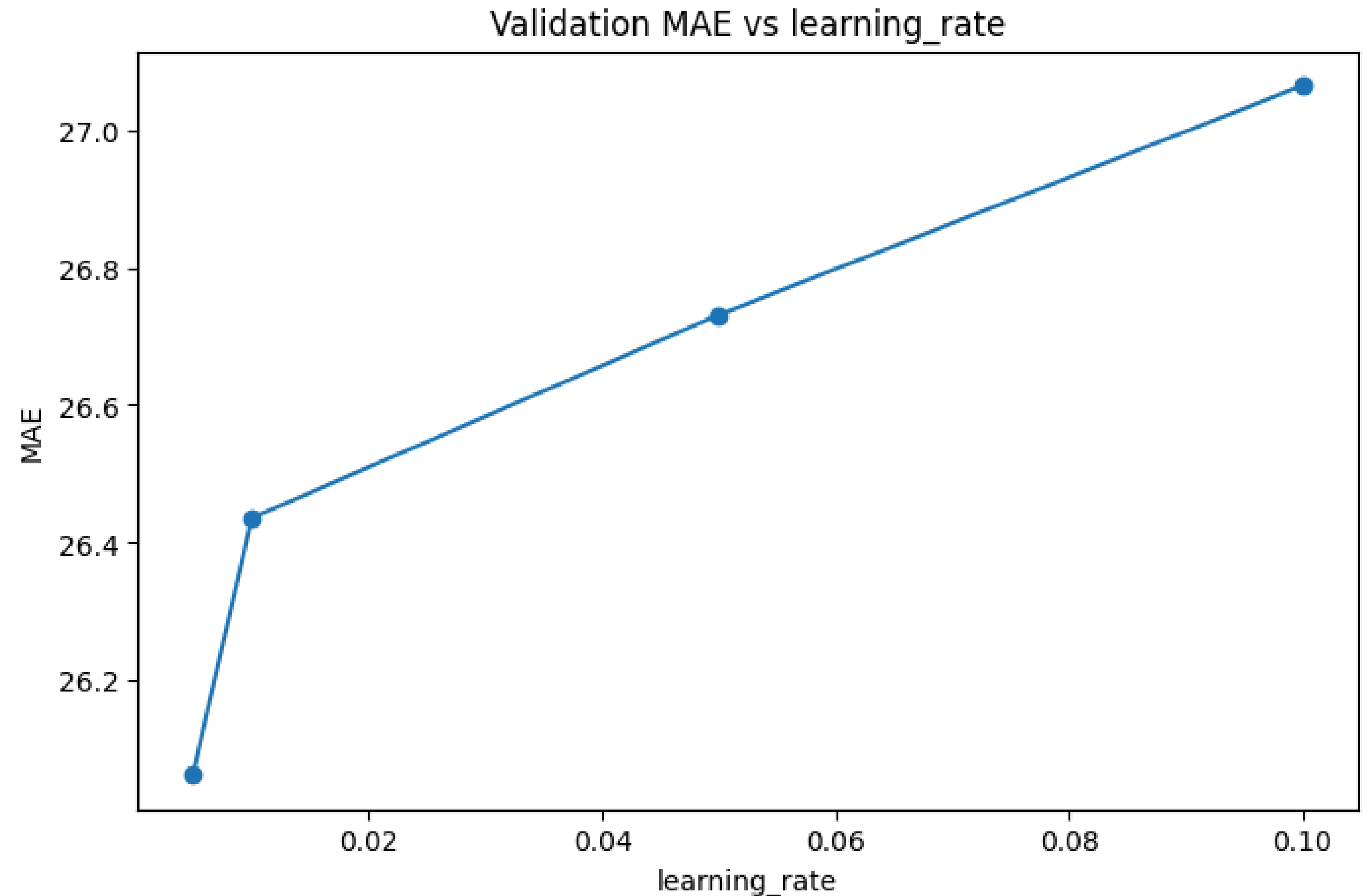


Hyperparameters Tuning

on **XGBoost** (with the previous preprocessing)

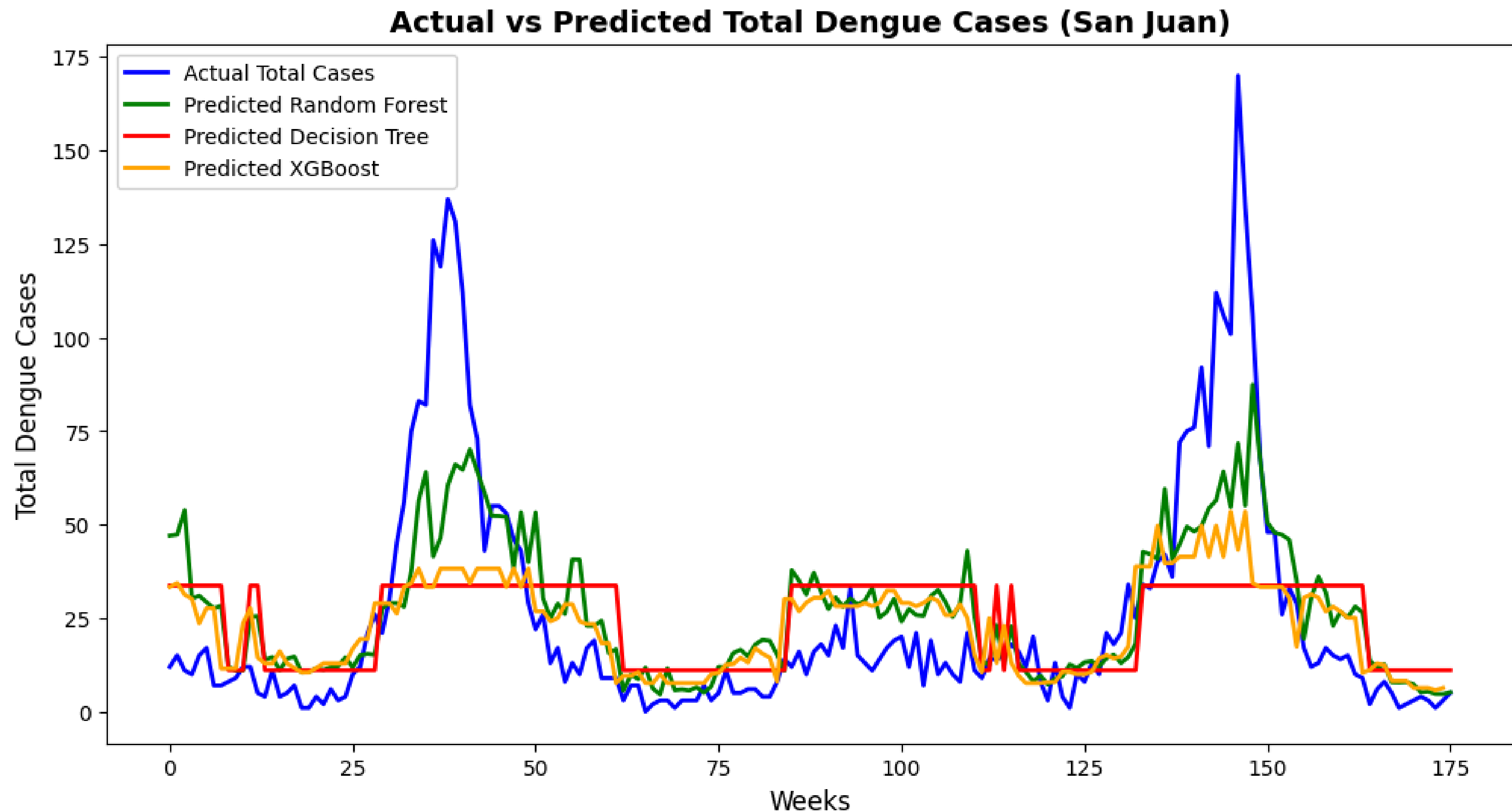
Best:

- learning_rate: **0.1**
- max_depth: **3**
- min_child_weight: **1**
- subsample: **1.0**
- colsample_bytree: **0.7**
- gamma: **3.0**



Final Model Comparison

RandomForestRegressor vs XGBoost vs SingleDecisionTree (Dummy Regressor)



Random Forest

MAE: **14.07**

RMSE: **21.36**

XGBoost

MAE: **15.53**

RMSE: **25.25**

Single Decision Tree

MAE: **18.74**

RMSE: **28.64**

Take-away message

- Lagged features greatly improves the model by identifying significant changes.
- log-Transform helps the model avoid overestimating by normalizing the distribution.
- Added features help the model predict better during outbreaks.
- Feature selection gets rid of correlated features.
- Random forests and XGBoost can predict the total dengue cases, but accurately predicting the outbreaks is challenging.

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